

1. Is C# case sensitive when dealing with identifiers?

☒ A	Yes	☐ B	No
☐ C	machine dependent	☐ D	none of the mentioned

2. In a C# program, a control structure:

☒ A	Directs the order of execution of the statements in the program	☐ B	Dictates what happens before the program starts and after it terminates
☐ C	Defines program-specific data structures	☐ D	Manages the input and output of control characters

3. What signifies the end of a statement block in C#?

☐ A	A line that is indented less than the previous line	☒ B	}
☐ C	A comment	☐ D	end

4. In C#, a variable may be assigned a value of one type, and then later assigned a value of a different type:

☒ A	true	☐ B	false
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5. What is the value of the expression $1 + 2 \wedge 3 * 4$?

☐ A	4097	☐ B	108
☐ C	36	☒ D	33

6. What is the output of the following program?

```
i = 1
while (True){
    if(i%3 == 0)
        break;
    console.WriteLine(i);
    i += 1; }
```

☐ A	No Output	☒ B	Error!
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7. The C# interpreter takes the code that you write and converts it to the language that the computer's hardware understands. Is this statement True or False?

☒ A	True	☐ B	False
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8. Suppose you have the following variables defined:

$a = 100$

$b = 50$

Write a C# if/else statement to assign variable m to the smaller of a and b .

☒ A	if ($a < b$) $a = m$; else $b = m$;	☐ B	if ($a < b$) $m = a$; else: $m = b$;
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9. In C#, a variable must be declared before it is assigned a value:

☒ A	true	☐ B	false
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10. Which is the special symbol used in C# to add comments?

☐ A	\$	☒ B	//
☐ C	#	☐ D	//*... */

11. In C#, strings are...

☒ A	immutable	☐ B	char arrays
☐ C	changeable	☐ D	str objects

12. Which of the following mathematical operators can be used to concatenate strings:

☐ A	/	☐ B	*
☒ C	+	☐ D	-

13. In the C# statement $x = a + 5 - b$:

a and b are _____

$a + 5 - b$ is _____

☐ A	terms, a group	☐ B	operators, a statement
☒ C	operands, an expression	☐ D	operands, an equation

14. Write the output of the following program:

```
static void Main (){
    unsigned int i = 10;
    while (i-- >= 0)
        cout << i << endl;
    system("pause");
    return 0;
}
```

☐ A	10 9 8 7 6 5 4 3 2 1 0	☒ B	infinte loop
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15. Write the output of the following program:

```
static void Main(string[] args){
    int a, b, c;
    for (a = 0; a < 1; a++)
        for (b = 0; b < 3; b++)
            for (c = 0; c < 3; c++)
                if (a != b && a != c && b != c) Console.WriteLine ("0{1}{2}", a , b , c);
}
```

☒ A	012 021	☐ B	012021
------------------------------------	------------	-------------------------	--------

16. Write the output of the following program:

```
static void Main () { int x=7,y=8;
    Console.WriteLine((x & y));
    if((x + y) == (x | y))
        Console.WriteLine("Not OK");
    else
        Console.WriteLine("OK");return 0;}
```

☒ A	0 Not OK	☐ B	0 OK
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17. Write the output of the following program:

```
static void Main () {int a = 12, b = 12, c = 1;
    c = (a - b) > 0 ? 10 : 12;
    Console.WriteLine("The value of c is: {0}", c);}
```

☐ A	The value of c is: 10	☒ B	The value of c is: 12
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18. Write the output of the following program:

```
static void Main (){
    int a = 5, b = 6, c = 7;
    if (a > b)
        if (b < c)
            c = a;
    else
```

$c = b;$

`Console.WriteLine("c= {0}",c); }`

A	C=7	B	C=6
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19. Write the output of the following program:

```
static void Main () {  
    int number = 1;  
    while (number <= 3) {  
        Console.WriteLine("{0} squared is {1}", number, SQR(++number));  
    }  
    static int SQR(int x) { return (x) * (x); }  
}
```

A	2 squared is 4 3 squared is 9 4 squared is 16	B	1 squared is 4 2 squared is 9 3 squared is 16
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20. Write the output of the following program:

```
static void Main () {  
    int number = 1;  
    while (number <= 3) {  
        Console.WriteLine("{0} squared is {1}", number, SQR((number++)));  
    }  
    static int SQR(int x) { return (x) * (x); }  
}
```

A	1 squared is 1 2 squared is 4 3 squared is 9	B	0 squared is 1 1 squared is 4 2 squared is 9
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21. Write the output of the following program:

```
static void Main(string[] args) {  
    int[] arr = new int[10];  
    arrayMethod(ref arr);  
    Console.WriteLine(arr.Length);  
}  
static void arrayMethod(ref int[] a){  
    int[] b = new int[5];  
    a = b;  
}
```

A	5	B	10
----------	---	----------	----

22. Write the output of the following program:

```
static void Main(string[] args) {  
    char[] asd = { 'A', 'B', 'C' };  
    string txt = "";  
    for (int a = 0; a < 3; a++)  
        for (int b = 0; b < 3; b++)  
            for (int c = 0; c < 3; c++)  
                if (a != b && a != c && b != c)  
                    txt += asd[a].ToString() + asd[b].ToString() + asd[c].ToString();  
    Console.WriteLine(txt);  
}
```

A	ABCACBBACBCACBACAB	B	ABCACBBACBCACABCB
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23. Write the output of the following program:

```
static void Main(string[] args) {  
    int[] arr = new int[] { 1, 2, 3, 4, 5 };  
    Console.WriteLine(fun1(ref arr));  
}
```

```

}
static string fun1(ref int[] array) {
    for (int i = 0; i < array.Length; i = i + 2)
        array[i] = array[i] + 10;
    return (string.Join(", ", array));
}

```

A	11,2,13,4,15	B	11,2,12,4,16
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24. Write the output of the following program:

```

static void Main(string[] args){
    int[] i = new int[0];
    Console.WriteLine(i[0]);
}

```

A	0	B	Exception
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25. Write the output of the following program:

```

static void Main(string[] args) {
    var stringValue = "Hello World!";
    var stringValue2 = stringValue;
    stringValue = "Hello Egypt!";
    Console.WriteLine(stringValue2);
    var array = new[] { 1, 2, 3, 4 };
    var array2 = array;
    array[0] = 99;
    Console.WriteLine(array2[0]);
}

```

A	Hello World! 99	B	Hello Egypt! 99
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26. Write the output of the following program:

```

static int div(int a, int b = 5){
    return (a + b);
}

static void Main(string[] args){
    Console.WriteLine(div(120));
    Console.WriteLine(div(120, 4));
}

```

A	124 125	B	125 124
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27. Write the output of the following program:

```

static void Main(string[] args){
    var v = "Hello World!";
    v = 10;
    System.Console.WriteLine(v);
}

```

A	Compile error	B	Run time error
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28. Write the output of the following program:

```

static void Main(string[] args){
    int[] Asd = { 1, 8, 3, 6, 2, 5, 9, 3, 0, 2 };
    Value(Asd, ref Asd[0]);
    Console.WriteLine("The return value is {0}", Asd[0]);
}

static void Value(int[] asd, ref int Val){

```

```

for (int i = 1; i < asd.Length; i++) {
    if (asd[i] > Val)
        Val = asd[i];
}

```

A	The return value is 9	B	The return value is 1
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29. Write the output of the following program:

```

static void Main(string[] args){
    int[] S = { 1, 2, 3, 4 };
    int[,] R = { { 1, 2 }, { 3, 4 } };
    int[][] J = { new int[] { 1, 2 }, new int[] { 3, 4 } };
    Console.WriteLine("{0} {1} {2}", S.Length, R.Length, J.Length);
}

```

A	4 4 2	B	4 4 4
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30. Write the output of the following program:

```

static void Main(string[] args){
    class A a = new class C();
    Console.WriteLine(a.Print());
}
public class class A{
    public virtual string Print(){
        return "class A";
    }
}
public class class B : class A {
    public override string Print()
    {
        return "class B";
    }
}
public class class C : class B{
    public new string Print(){
        return "Class C";
    }
}

```

A	Class C	B	Class B
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31. Which of the following statements are correct for the given code snippet:

shape obj;
obj = new shape();

A	Creates an object of class shape.	B	To create an object of type shape on the heap or stack depending on its size.
C	Create a reference obj of the class shape and an object of type shape on the heap.	D	Create an object of type shape on the stack.

32. Write the output of the following program when the input is 5:

```

static void Main(string[] args){
    int N = int.Parse(Console.ReadLine());
    int first = 0, second = 1, next;
    for (int i = 0; i < N; i++){
        if (i <= 1)
            next = i;
        else{
            next = first + second;
        }
    }
}

```

```

        first = second;
        second = next;
    }
    Console.WriteLine("{0} ", next);
}
}

```

A	1 1 2 3 5	B	0 1 1 2 3
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33. Write the output of the following program when the input is 5:

```

static void Main(string[] args){
    int n = int.Parse(Console.ReadLine());
    for (int i = 1; i <= n; i++){
        for (int j = 1; j <= i; j++){
            Console.Write("{0}{1}", j, " ");
        }
        Console.WriteLine();
    }
}

```

A	1 1 2 1 2 3 1 2 3 4 1 2 3 4 5	B	1 1 3 1 3 5 1 3 5 7 1 3 5 7 9
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34. Write the output of the following program when the input is 5111:

```

static void Main(string[] args){
    int N = int.Parse(Console.ReadLine());
    int sum = 0;
    while (N != 0){
        sum += N % 10;
        N = N / 10;
    }
    Console.WriteLine(sum);
}

```

A	88	B	8
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35. Write the output of the following program:

```

static void Main(string[] args)
{
    Console.WriteLine ("*\n**\n***\n****\n*****\n");
}

```

A	* ** *** **** *****	B	***** **** *** ** *
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Good Luck , Prof. Yousef Bassyouni Mahdy